EP491 Project Proposal

Ethan Bessette

https://docs.google.com/document/d/1AQb_9IuocH631eH7uKp3l9mJWkfTmy0e_qxKSp5Wgos/edit?usp=sharing

Project Overview

What is the project about? What is the final outcome? What is the presentation format? Why is it important to me? How will it demonstrate my learning in EPD? How does it connect to my future goals?

This project is a multimodal interactive experience, or installation, that aims to connect the kinetic with the visual and auditory modes using the gestures and positions of humans to drive a procedurally generated visual system. The experience will involve the exploration of potential gesture combinations, such as in online particle generators where combining particles creates new types of particles. It will also involve the exploration of a virtual space as the participants progress through a short story. The final outcome will be a binary executable and production documentation, and the presentation format will be an installation work that runs the executable using a computer, a projector or screen, and audio amplifiers. It will also be recorded as a video to be more accessible as part of a digital portfolio. This project is important to me because it combines my education in interactive audio system creation with my interests in sound design and programming. I also personally love finding new sounds and textures, and this experience will be a structured way to share that joy of exploration with participants by allowing them to use their bodies to create new combinations. For some idea of what it will look like, view the Storyboard Rough Draft (see below).

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Required Tools

Sensor: a way to convert body movement in space into a digital signal

- Could be a visual system like a camera, stereo camera, or lidar
- Could be a location tracking system like a VR headset and controllers
- I could use a **sensor** paired with the **Arduino** that I have and connect this to a laptop
- I could use the **built in camera** on a computer or phone

Software to recognize specific movements from sensor data

- For example, hand movement versus head position to track when hands above head, or when a swimming motion is made
- I could use **Wekinator** and train the events I want it to recognize based on the sensor data. It would then output messages when events are recognized, no matter the speed in which they are performed. **Dynamic Time Warping** (**DTW)**
- I could use **this** **Unity package that supports DTW**
- I could use the Unity package **Sentis/Inference** to import ML models
- Look into **MediaPipe** and **YOLO**

Software to run the experience

- **Unity**. Version TBD based on package dependencies such as listed above
- **Audio middleware** like **Wwise**

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    - Consider [this] AudioKinetic blog post about using RTPCs to modify Unity objects
        - Use filters to isolate frequencies, use meter after filters to send to RTPC, create timbre dependent mapping by storing patterns of levels of the dif freq bands (ex: if band4 amplitude is in this range and band 2 is in this range, do something)
        - could create light intensity and color changes based on both game state/gesture recognition, and realtime audio timbre and intensity
- **Visual creation** possibly using **ShaderGraph**
- Look into this Unity package that can map audio data to shaders and unity visual systems like lights. It doesn't work with Wwise, but is there anything it has already set up, especially for converting to shader data, that I could pull from for using Wwise? For example, it already has a script to map audio data to control light color and intensity. Could I instead replace audio data input in the script with RTPC value?

AI coding agent: possibly. Thorough documentation of prompts and responses would be kept if used.

Computer to run binary distribution during showcase**: Macbook M4 Pro**

Projector or screen to show visuals during showcase

Amplification to play audio during showcase

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Required Collaborators

While most of the work will be self directed, the following is a list of potential roles as needed: writer / storyboarder, 3D visual designer, level designer, programmer, conductor, recording engineer.

12-Week Production Timeline + Pre-semester Prep

Preproduction

- [ ] Decide on hardware setup
- [ ] Prototype basic movement tracking in Unity; determine model to use
- [ ] Find a writer / artist who is interested in doing storyboarding and concept art for the project
- [ ] Create spreadsheet for mapping gestures to visuals and audio descriptions
- [ ] Draft GDD and audio style guide doc

Week 1-2: Project Structure

- [ ] Build initial Unity project structure (see below)
- [ ] Draft of audio due for initial storyboard (see below)

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Week 3-8: Agile Loop

- [ ] Main agile loop (see below)

Week 9: Asset Refinement and Buffer

- [ ] Wrap up the current agile loop. Respond to feedback and refine assets

Week 10: Testing and Mixing

- [ ] Run tests, adjust based on feedback
    - Responsiveness
    - Clarity of interactions
    - Feedback on audiovisuals
    - General feedback
- [ ] 1-2 mix sessions

Week 11: Refinement and Mixing

- [ ] Polish audio transitions and mix (1-2 mix sessions)
- [ ] Optimize visuals and performance
- [ ] Prepare installation setup

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Week 12: Final Prep

- [ ] Set up presentation space
- [ ] Record demo video
- [ ] Finalize written documentation
- [ ] Celebrate!!

Build initial project structure

1. The input system will be based on **gesture events**
    a. Similar to how specific keys are the input events in normal games
2. **Larger game states**
    a. possibly based on location, story, pause, etc
    b. Allow subsystems to do different things based on this state
3. Handle audio **larger game state** switches in Wwise
    a. or possibly also with a **strategy design pattern** to limit direct calls to Wwise audio engine and to set up singletons for each event that shouldn't be retriggered like background ambience. Even though Wwise has internal voice limits it still sometimes has problems and abstracting Wwise events as scriptable objects could be a good solution
    b. Use Wwise switch groups and/or strategies so that the same **gesture events** trigger different **audio events** based on game state

4. Handle visual **larger game state** switches with a custom class
    a. **Strategy design pattern**
        i. The same **gesture events** trigger different **visual events** based on game state
5. **Gesture events** will control the player like normal input events, triggering **player state changes**
    a. Idle state: allows visualization of arm movement
    b. Swim state: virtual movement, trigger **visual events** and **Wwise events**
    c. Use state factory
6. Overtime the player might gain **player abilities**
    a. **Strategy design pattern**
        i. Locomotion ability: swim **gesture event** triggers swim **player state change**
        ii. No Locomotion ability: swim **gesture event** does nothing

Main Agile Loop

1. Planning
    a. Define a few **gesture events**
    b. Map **gesture events** and combinations of them to and write descriptions for:
        i. **Audio events**

ii. **Visual events**
iii. **State changes**

c. Write descriptions of transitions between events, states
d. Additionally, plan how shorter term audio data like amplitude and frequency distribution might map to the visuals
e. Determine how these gestures will contribute to the story or experience.
   i. Initial gestures: idle, arm height, locomotion, participant proximity

2. Create audio and visual assets for the different **gesture events**, **game states**, **player states** and transitions from the planning phase
3. Add the gestures to the Unity Project structure
   a. Test to make sure states and abilities are being properly updated
4. Implement the audio and visual assets
5. Test. Do some refining of audiovisuals, gesture tracking. Especially transitions since it's the first time they can be fully heard. Keep notes for future refinements. Prioritize moving on if it functions, even if it's not perfect.
6. Optional: if another loop cannot be completed in time, perfect and refine assets based on notes from the test phase.

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Storyboard Rough Draft

Lydian Dorian major/minor scale

Upon start, pitch black visuals with very slight grey dots and streamers. Vaguely muffled watery moving ambience

When person first steps in front of kinnect (or first steps forward in VR) a glowing vaguely human shaped ball of light enters the screen and a deep string chords plays mixed with warm ambient synth.

Generated synth tremolo effect like the wii menu but more shimmery and less transient and more ambient is underneath full the whole time. Music slowly fades to a dull ember if arms not put up. Strings a dark but warm sound.

Transition either from start state or arms up state. pulses/swells, make it dynamic moving. Bass moves with chords swelling above.

Either transition from start or arms down state. Maybe a masking noise for transition. The glowing orb sprouts a warm colored arm in sync with player arm, and triggers a scale upwards from whatever bass note was playing. The strings fade in to play a brighter higher chord (still warm and not blinding, just lighter). Chord progression starts climbs to a predefined point. If arms stay up to reach that point.

If stay up, strings reach a climax and subside into that synth tremolo and also sparkles like wind chimes, layered with one high string playing generated notes from a collection with a distribution map for what notes can go where based on previous note. Plays a certain number before fading to just the shimmery effect, either ending in highest or if it goes down 3 before (so towards end of predefined number of notes it either has to continue climb or go down). And when arms down last note played rings out and the shimmery effect subside back to dull lower pitch behind swells, return to base state.

Else if arm goes down before reaches that point in the ascending scale, scale steps back down 3 steps and back up 1 step and hangs that chord till fading back to arms down, unless arms up before it fades which starts climbing again from that chord.

Create transitions from every state to this one when 2 arms are up. The blob now has 2 glowing arms up. Light bell ring, Even bigger swell with faster string scales above slower moving ones, swell in harp, as the area lightens up and you start to see lights and objects floating around you. At this point putting arms down propels you forward like swimming and doesn't immediately react in the audio till after this transition happens. Or if arms aren't down by the end of the transition, the game gives a hint by moving the arms of the character for the player, showing that moving arms in swimming motion propels forward and is audio reactive.

The ambience changes to reflect the lighter area

Thanks for reading! :))